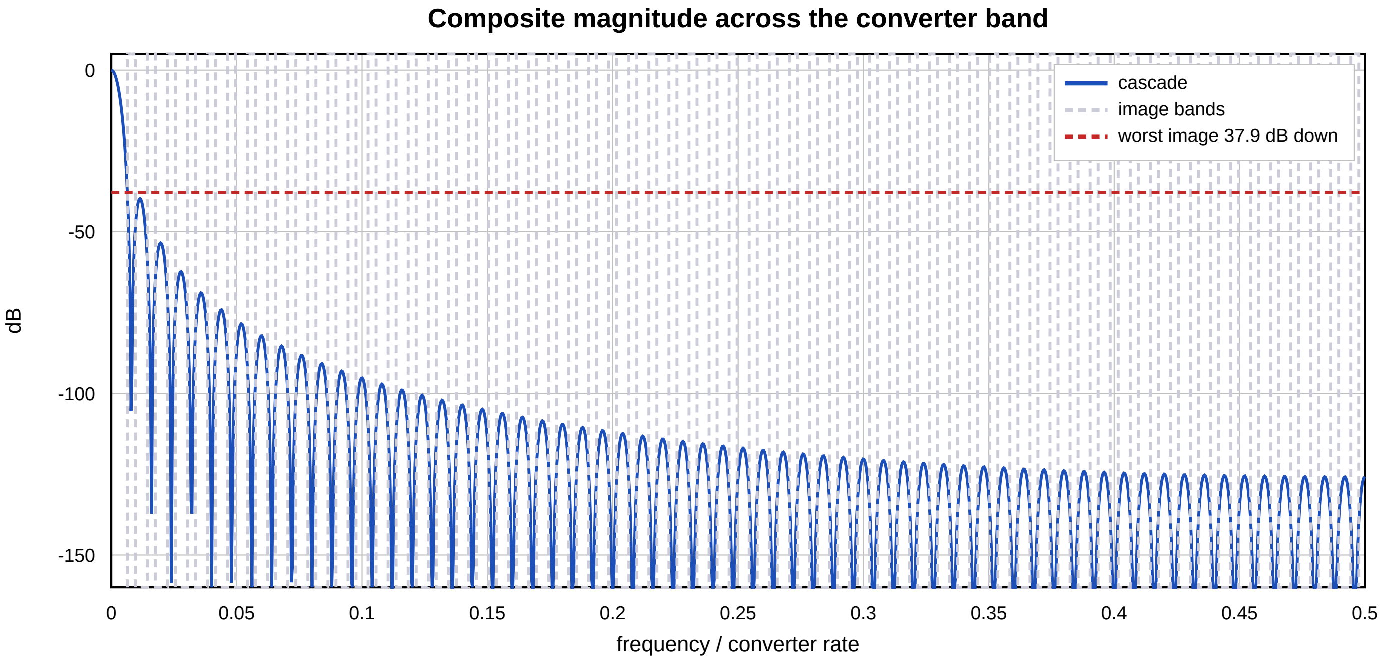
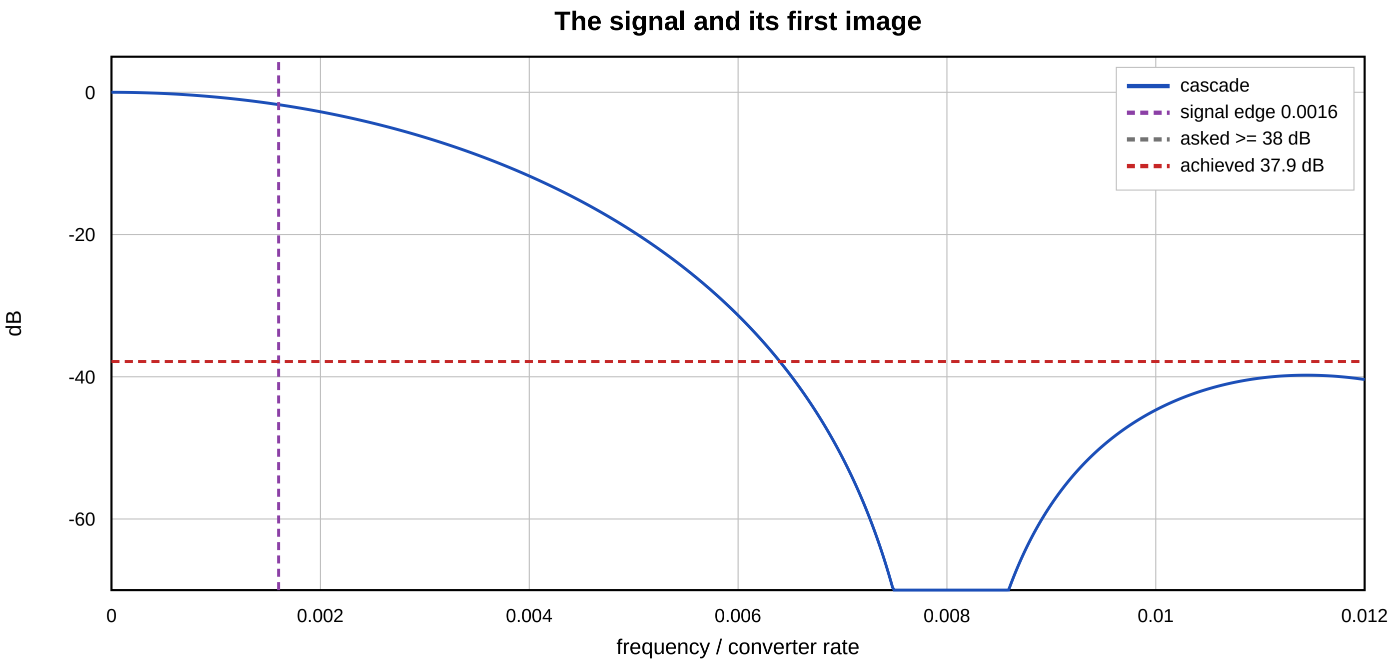
Interpolation Chain - Specified Parameters

1000.0000 kHz x 125 = 125.000 MHz, split [125], 200.000 kHz signal





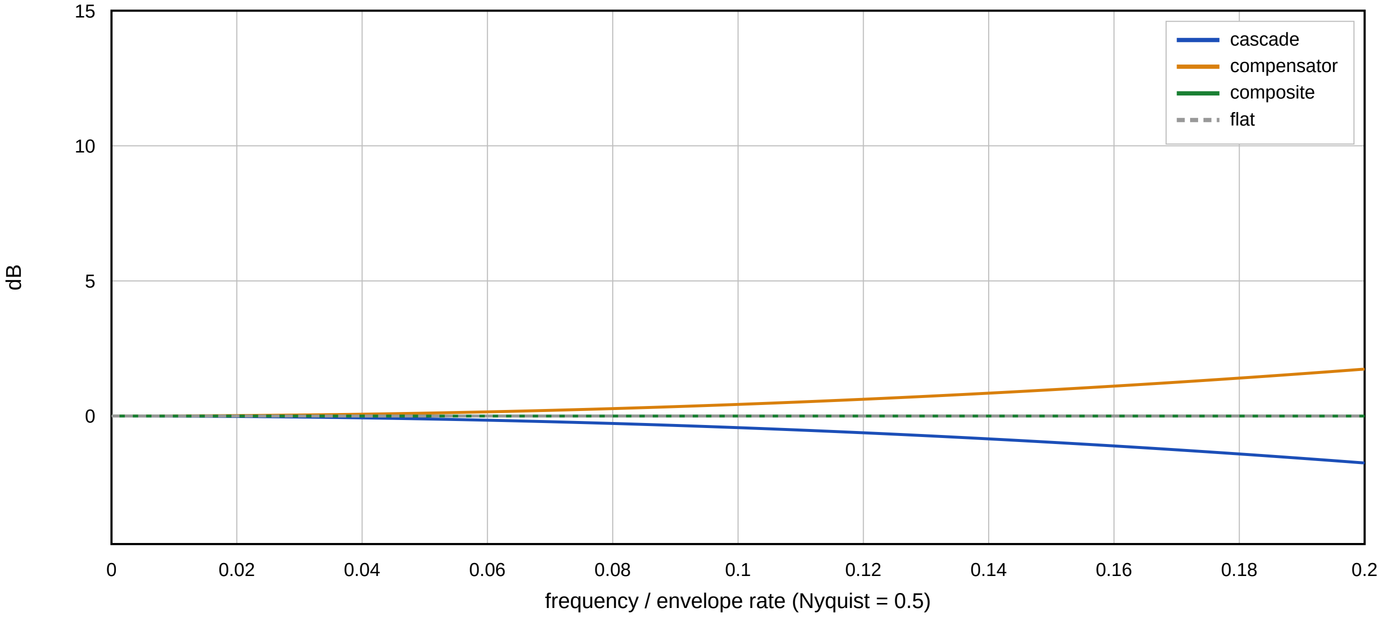
Stages

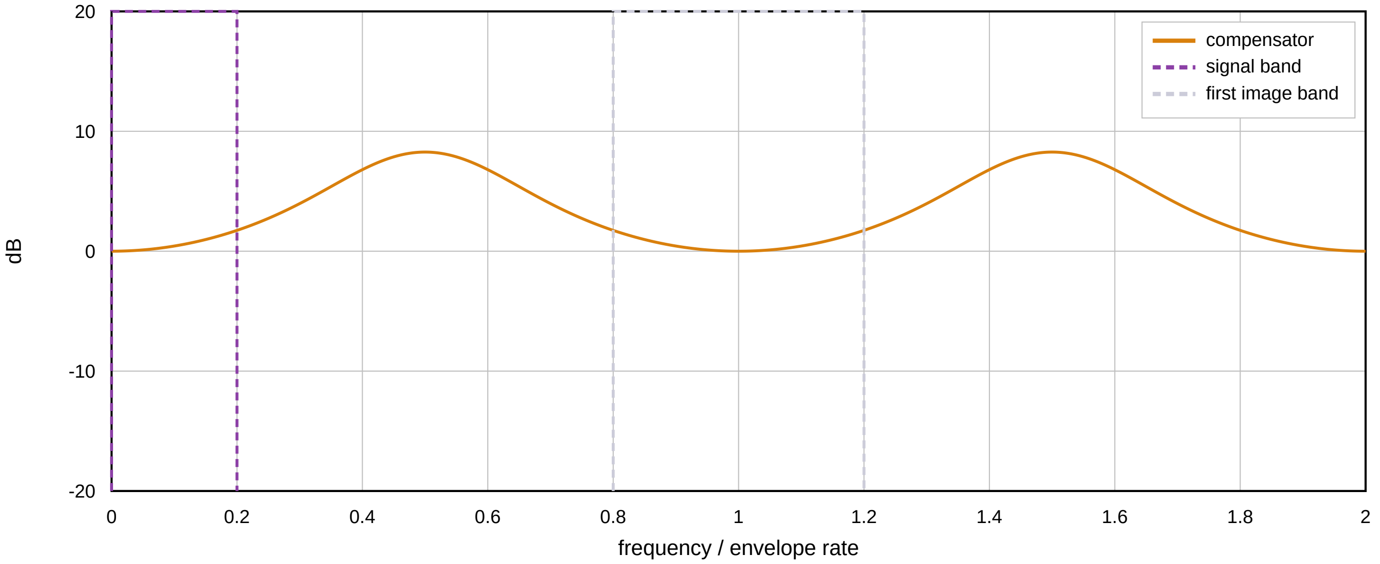
stage 1: x125 N=3 M=1 combs at 1.0000 MHz, integrators at 125.000 MHz
in 16 bits, uniform accumulator 30 bits, tapered widths [17, 18, 19, 18, 24, 30]
270 register bits uniform, 181 as built (lossless), bound 180
widths as built [17, 18, 19, 19, 24, 30]

The combs run at the stage's input rate and the integrators at its output rate, so an integrator bit costs R times what a comb bit costs. Depth belongs early and rate late. The tapered widths are lossless: each stage sized to its own growth bound holds its value exactly, so a tapered interpolator is bit-identical to a uniform one.

Compensation and Result

15 taps, 16-bit coefficients, at the 1.0000 MHz envelope rate





Figures

images .... 37.9 dB down (asked >= 37.9)
   worst image at .... 0.8000 MHz
passband ripple ...... 0.0011 dB (asked <= 0.001)
uncompensated droop .. -1.738 dB at the band edge
converter floor ...... 86.0 dB at 14 bits (the chain adds none)
register bits .... 270 uniform, 181 as built, 180 at the exact bound
compensator DC gain .. 1.0000, shift 14
compensator headroom . 18 bits for any input, 17 for in-band only
   norms .... l1 2.5906, passband peak 1.2214
adder depth .... 3 deep in the combs, 1 in the integrators

taps .... [0, 4, -27, 130, -506, 1737, -5982, 25672, -5982, 1737, -506, 130, -27, 4, 0]

Build to the any-input headroom unless the envelope is band-limited.

The in-band figure is the peak gain an in-band sinusoid sees. The any-input figure is the sum of the taps' magnitudes, which is the bound for an arbitrary bounded input -- and a transmit envelope is not band-limited: switch-on, a burst boundary and a modulation change are all steps, and a step is exactly the input that reaches it. Choosing the narrower figure for a burst transmitter is how a compensator saturates on the first sample.

More taps will not improve image rejection.

The compensator runs before the rate change, at the envelope rate, so its response is periodic and the image at k+u sees exactly the gain the signal at u sees -- the plot above shows it. The image-to-signal ratio is the cascade's alone. This is the one place where